

4 November 2019

Number:

Name Surname:

**CEN 209 Differential Equations
QUESTIONS**

- 1.** Find the general solution of the differential equation

$$\left(\frac{y}{x} + \sec^2 x\right)dx + (2y + \ln x)dy = 0.$$

- 2.** Find the general solution of the differential equation

$$\frac{dy}{dx} + \left(\frac{x+1}{2x}\right)y = \frac{x+1}{xy}.$$

- 3.** Find the general solution of the differential equation

$$\left(x \tan\left(\frac{y}{x}\right) + y\right)dx - xdy = 0.$$

- 4.** Given that $y = \frac{1}{x}$ is a solution of

$$x^2 \frac{d^2 y}{dx^2} - x \frac{dy}{dx} - 3y = 0.$$

Then find the general solution of the differential equation.

Time: 75 minutes.

Best Luck